User Manual

Low-Noise Wideband Amplifier

LNA-03-30-A-R



User Manual – Version v1.2.5
06 Nov 2026

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1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the Low-Noise Wideband Amplifier LNA-03-30-A-R.

This document forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

1.1.2 Revision History

Document Revision: v1.2.5- 06 Nov 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the device.

Provides important supplementary information.

Indicates procedures that, if not observed, may result in damage to the device or associated equipment.

Indicates procedures that, if not observed, may result in injury or serious damage.



(a) General hazard symbol



(b) Mandatory reading of the manual

Figure 1.1: The device may include the pictograms shown above.

All safety symbols must remain clearly visible throughout the lifetime of the device.

1.3 Manufacturer Information

Manufacturer: FLIM LABS S.r.l.
Address: Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone: +39-329 761540
E-mail: info@flimlabs.com
Website: www.flimlabs.com

1.4 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.2. Disposal requirements associated with the WEEE marking are described in Section 7.2.

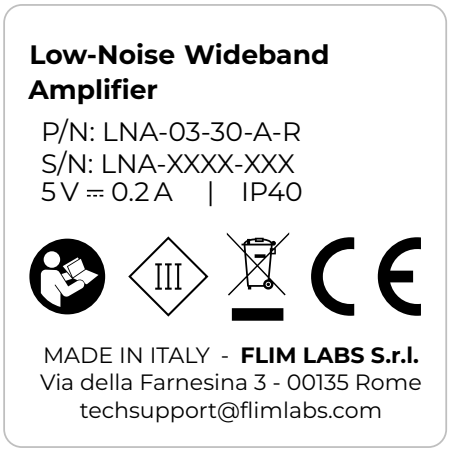


Figure 1.2: Product label

If the label becomes illegible or damaged, contact authorized service for replacement.

Safety pictograms performing a protective function must not be removed, covered, or damaged.

1.5 Declaration of Conformity

The device described in this documentation is sold with a Declaration of Conformity drawn up in accordance with applicable European legislation. Before using the device, verify the presence of the Declaration of Conformity. A copy of the Declaration of Conformity is shown in Figure 1.3.

EU Declaration of Conformity

CE

Device Information	
Type:	Low-Noise Wideband Amplifier
Model:	Low-Noise Wideband Amplifier
Part Number:	LNA-03-30-A-R
Serial Number:	As indicated on the product label

Manufacturer Information	
Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 ROME (RM), Italy
Telephone:	+39 – 329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

Applicable Directives and Regulations

- Directive 2014/35/EU, known as the “Low Voltage Directive”
- Directive 2014/30/EU, known as the “Electromagnetic Compatibility Directive”
- Directive 2011/65/EU, known as “RoHS”
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as “WEEE”
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

Place and Date	Signature
Rome, 24/10/2025	Alessandro Rossetta Alessandro Rossetta, CEO

Figure 1.3: EU Declaration of Conformity

1.6 Warranty and Liability

1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

1.6.2 Limitation of Liability

The device has been designed, manufactured, and tested in accordance with applicable European directives and internal quality standards. The manufacturer is liable exclusively for defects attributable to manufacturing faults or non-conformity with declared specifications, within the limits defined in the warranty terms.

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

2 Device Description

This section describes the functional characteristics, intended application, and technical features of the Low-Noise Wideband Amplifier.

2.1 Functional Description

The low-noise broadband amplifier is a compact electronic module designed for amplification of high-frequency electronic signals, in particular low-amplitude pulsed signals in the millivolt range used in scientific instrumentation, photon counting, and fluorescence lifetime imaging microscopy (FLIM) applications.

The device receives electronic input signals from external devices such as single-photon detectors, discrimination modules, or other front-end electronics, and provides an amplified output signal compatible with downstream acquisition, timing, and analysis instruments. Input and output signals are connected via standard SMA coaxial connectors suitable for high-speed, low-distortion signal transmission.

The module features:

- Compact electronic amplifier module
- One SMA input for the signal to be amplified
- One SMA output for the amplified signal
- Non-inverting amplification
- Broadband operation up to the specified cut-off frequency
- Low timing variation
- Compact aluminum enclosure suitable for system integration
- USB Type-C power supply
- Power status LED

2.2 Intended Use

The device is intended as a compact electronic module for amplification of electronic signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone unit or integrated into custom measurement systems.

2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

2.4 Components Overview

Figure 2.1 shows the key components of the device, including:

1. USB Type-C power input
2. SMA signal input
3. SMA amplified signal output
4. LED power indicator
5. Mounting holes in the enclosure

2.5 Mechanical Drawings and Dimensions

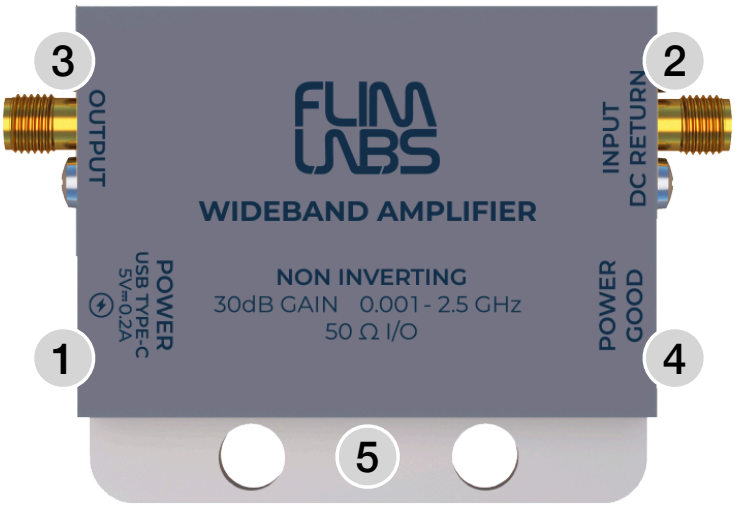


Figure 2.1: Top view showing the key components of the Low-Noise Wideband Amplifier.

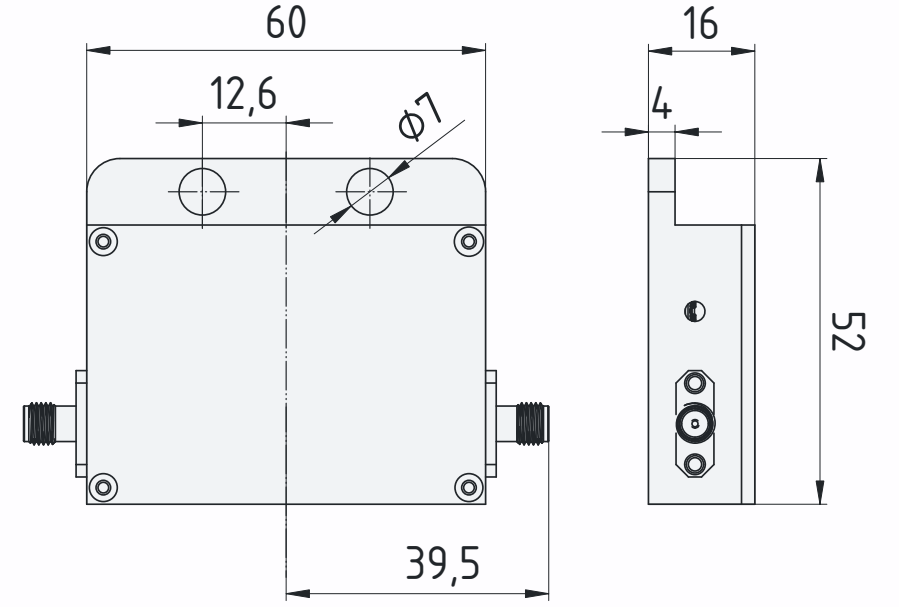


Figure 2.2: Technical drawing of the Low-Noise Wideband Amplifier.

2.6 Technical Specifications

The electrical, signal, and environmental limits referenced throughout this manual are summarized in this section.

Item	LNA-03-30-A-R
Electrical Specifications	
Electrical safety class	Class III
Power supply	USB Type-C, 5 V DC
Current consumption (max)	0.2 A
Power consumption (max)	1 W
Signal Specifications	
Input connections	1x SMA
Input polarity	positive or negative
Maximum input amplitude	5 V
Output connections	1x SMA
Voltage gain	30 dB (>30x)
Cut-off frequency (-3 dB)	2.5 GHz
Low-frequency limit	1 MHz
Output P1dB	>4 dBm OP 1 dB
Amplification type	non-inverting
Timing variation	<5 ps
Impedance	50 Ω
Physical Specifications	
Dimensions (L x W x H)	60 mm x 42 mm x 14 mm
Weight	80 g
Protection rating	IP40
Environmental Specifications	
Operating temperature	0 °C to +40 °C
Storage temperature	-20 °C to +70 °C
Humidity	80% non-condensing

Table 2.1: Technical specifications of the Low-Noise Wideband Amplifier.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding.

The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Residual risks may remain if the device is not used as specified.

3.2 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

3.3 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

3.4 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

4.1 Scope of Delivery

- Low-noise broadband amplifier
- USB Type-C power cable
- Quick Start Guide
- Declaration of conformity

Verify that all items are present before proceeding with installation.

4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.
After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.
Dispose of packaging materials responsibly and keep them out of reach of children.
The device is compact and lightweight and may be moved manually. When handling, support it securely to prevent accidental dropping.

4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.3 are met. Children must not have access to the installation or storage area. The device requires a regulated 5 V DC power source via USB Type-C.



Before connecting the device, verify that the input signal characteristics are compatible with the technical specifications of the CFD (see Table 2.1). Improper signal conditions may result in malfunction or unreliable operation. An overview of the required input signal checks is provided in Figure 4.1.

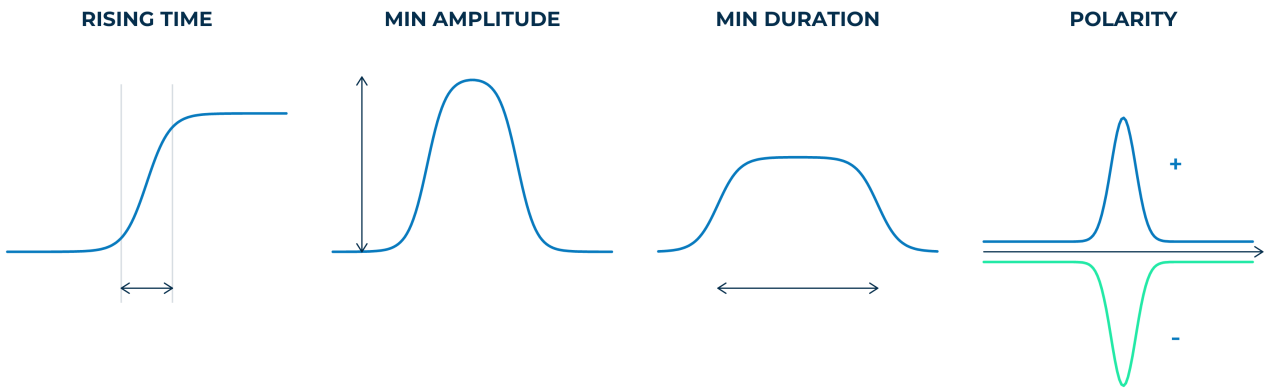


Figure 4.1: Before connecting any signal sources, verify that the input signal characteristics match the requirements in Table 2.1.

4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing and ventilation. Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The power supply location is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

4.5 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Connector locations are shown in Figure 2.1.
Proceed as follows:

1. **Power Connection**:
   a) Connect the USB Type-C cable to the device power input.
   b) Connect the other end to a suitable 5 V DC power source such as a computer USB port, power bank, or certified 5 V DC adapter.
   c) Verify visually that the LED power indicator confirms correct device power-up.
2. **Input Signal Connection**:
   a) Connect the external signal to be amplified to the SMA input connector.
3. **Output Signal Connection**:
   a) Connect the SMA output connector to the downstream acquisition, processing, or transmission instrument.

Example Connection Diagrams

Figures 4.2 and 4.3 show typical use cases for deploying an Low-Noise Wideband Amplifierin FLIM measurements. The components shown in the diagram perform the following functions:

Acquisition of Photomultiplier Signals

PMT The photomultiplier tube converts light into measurable electrical signals for fluorescence imaging.

Active Splitter Duplicates the PMT signal to integrate the FLIM upgrade while maintaining parallel acquisition by the microscope system.

Wideband Amplifier Amplifies weak fluorescence signals with minimal added noise, supporting accurate lifetime measurements.

Constant Fraction Discriminator Converts amplified PMT signals into photon-counting outputs, enabling the FLIM DAQ to record precise timestamps of single fluorescence photons.

Integration of a Laser Synchronization Signal

Excitation Laser Provides an electrical synchronization signal corresponding to the laser pulses via an external output port.

Wideband Amplifier Amplifies the laser synchronization signal with minimal added noise, supporting accurate detection of laser pulse timing.

Constant Fraction Discriminator Converts the amplified synchronization signal into digital timing pulses, providing the FLIM DAQ with a precise timing reference for fluorescence lifetime measurements.

The examples are provided for reference. The actual configuration may vary depending on the connected equipment and application.

4.6 Initial Configuration

After completing all electrical connections, no further configuration is required.

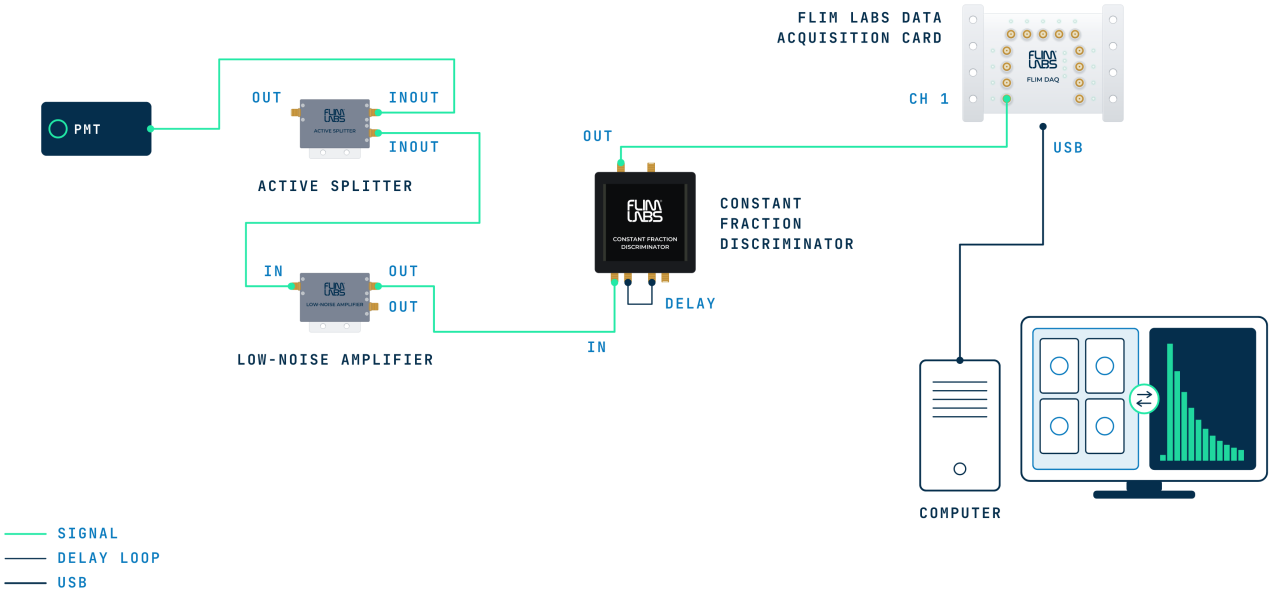


Figure 4.2: Example configuration connecting a PMT to a FLIM data acquisition channel.

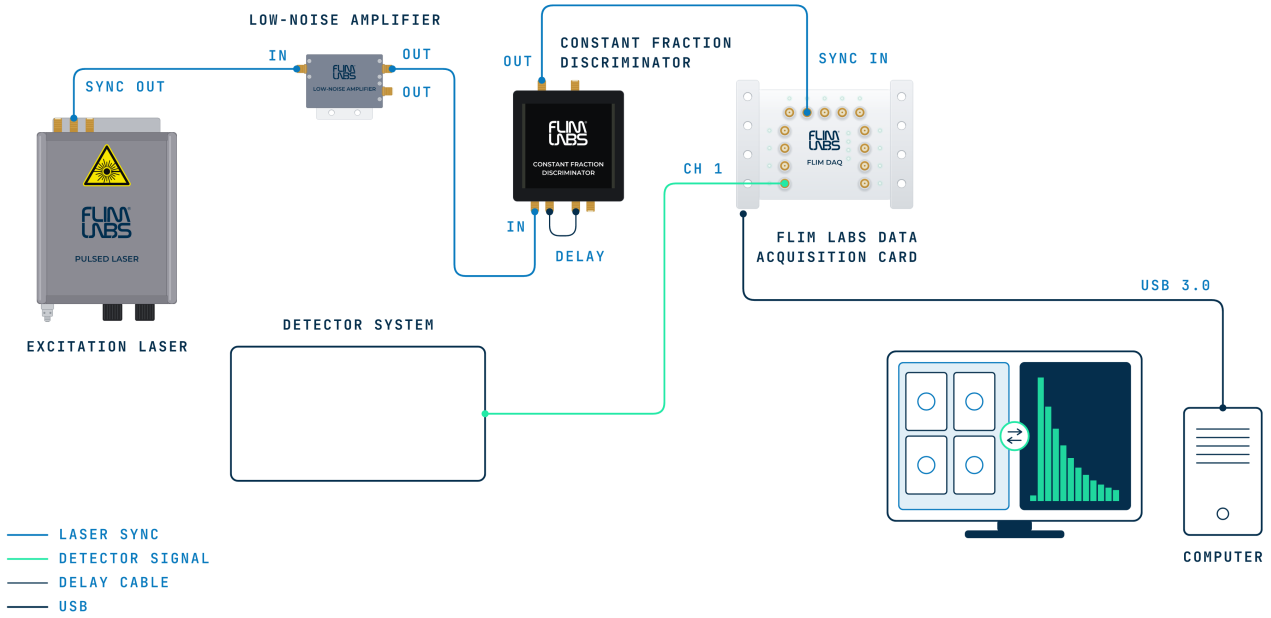


Figure 4.3: Example configuration using the LNA to connect a synchronization signal.

5 Operation

This section describes the functional behavior of the Low-Noise Wideband Amplifierduring use, including its operating principle, normal operating conditions, adjustable parameters, and troubleshooting guidance.

5.1 Principle of Operation

The amplifier receives an analog electrical input signal and provides a corresponding amplified analog output signal. The device operates as a non-inverting broadband amplifier intended to preserve signal integrity while increasing signal amplitude for subsequent acquisition, processing, or transmission.

Typical applications include amplification of signals from single-photon detectors, laser trigger signals, and other front-end electronics used in laboratory instrumentation.

5.2 Normal Operation



Do not operate the device if it is damaged or shows signs of unsafe condition.

Once the power supply, input signal, and output connections have been correctly established, the device operates autonomously without further user intervention. Correct functionality depends on proper system integration. The user is responsible for ensuring that the complete measurement setup meets applicable safety and performance requirements.

A typical example of the configured input and output waveforms is shown in Figure 5.1.

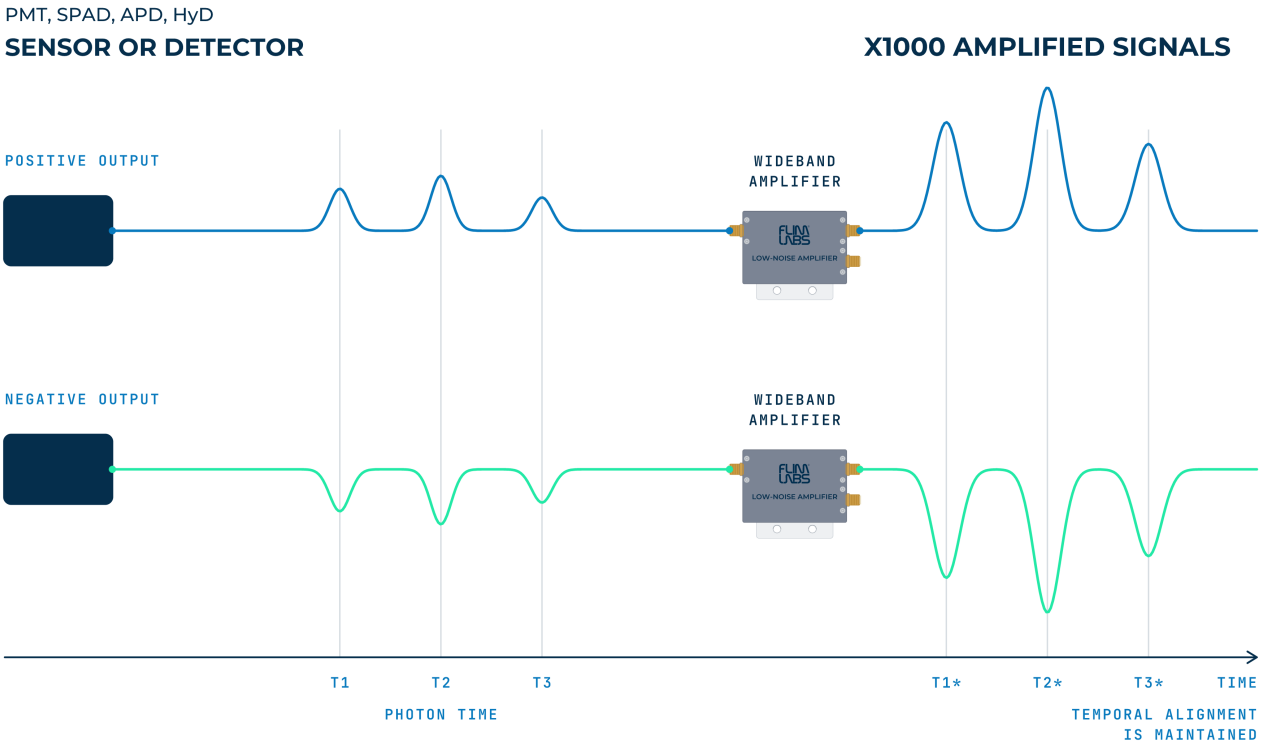


Figure 5.1: Schematic illustrating input and output signals of a LNA.

5.3 Parameter Adjustment

The amplifier does not require user-adjustable operating parameters for normal use.

5.4 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

Table 5.1: Overview of common problems and recommended corrective actions.

Problem	Possible Cause	Recommended Action
Power and Device Status		
Device does not power on	Insufficient USB power	Connect the device to a suitable 5 V DC USB power source. Avoid passive USB hubs and verify that the source can provide the required current.
Device does not power on	Defective USB cable or loose connection	Replace the USB Type-C cable and verify correct connector seating at both ends.
Power LED not illuminated	No power supplied to the device	Verify that the USB power source is active and that the device is correctly connected.
Input Signal		
No amplified output signal	No valid input signal present	Verify that an input signal is present at the SMA input and that the source device is operating correctly.
No amplified output signal	Input signal exceeds specified limits or is incompatible	Verify that the connected signal complies with the electrical specifications of the device, including maximum input amplitude and impedance.
Output signal unstable or noisy	Noise on input signal or poor grounding	Improve signal quality, reduce cable length, and ensure proper shielding and grounding.
Signal amplitude lower than expected	Input signal amplitude too low	Verify that the input source provides a sufficient signal level for the intended amplification chain.
Output Signal		
No signal detected by downstream instrument	Incorrect output connection	Verify that the SMA output is connected to the correct input of the downstream acquisition or measurement system.
Output signal distorted	Improper impedance matching or excessive cable length	Use short 50 Ω coaxial cables and ensure proper termination of the signal path.
Output signal not suitable for acquisition	Downstream instrument incompatible with amplifier output	Verify that the connected instrument is compatible with the amplified analog output signal and expected bandwidth.



If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.

Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

6.2 Preventive Maintenance

The device does not require routine internal maintenance.
Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation
- Product label for legibility. See Section 1.4 for label content and handling requirements.

If any abnormal condition is detected, discontinue use and contact technical support.

6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer’s technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.

Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.4). Contact your local waste management authority or certified collection center for proper disposal procedures.

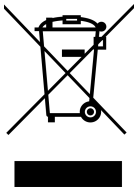


Figure 7.1: WEEE symbol